

RUT Derivation Target Note

Geometry-Light-Time Coupling v1 - upload note for GitHub / OSF

Purpose

This note preserves a central mathematical target for Reverse Universe Theory: eventually derive, or at least attempt to derive, a compact equation family showing how geometry, light-flow, and time/phase orientation are linked while preserving the GR/LambdaCDM correspondence limit.

This is not presented as a completed derivation. It is a timestamped derivation target: a clear statement of what RUT must eventually make precise.

1. Central principle

RUT's compressed target principle can be stated as follows:

geometry, light, and time are not independent primitives; they are three readable expressions of one deeper phase-order structure.

In this reading, geometry describes how structure curves and responds, light describes null/radiative direction and propagation, and time describes the oriented ordering of change. The RUT task is to show whether these can be written as one disciplined mathematical system rather than three separate ideas.

2. Candidate compressed equation family

A first target form can be written as an extended Einstein-type equation with ordinary matter plus three RUT-facing contributions: a phase/omega sector, a light-flow sector, and a mixed geometry-light-time coupling term.

$$G_{mn} = 8 \pi G [T_{mn}^m + T_{mn}^\omega + T_{mn}^L + \xi B(a) T_{mn}^{(\omega L)}]$$

Interpretation:

G_{mn} = geometry / curvature response; T_{mn}^m = ordinary matter sector; T_{mn}^ω = phase-order or cosmic-breath sector; T_{mn}^L = light-flow, radiation, or null-structure sector; $\xi B(a) T_{mn}^{(\omega L)}$ = the RUT bridge term coupling phase order and light-flow across cosmic epochs.

3. Light-time orientation law

A second target relation should connect time orientation to light-flow through the omega/phase field. A cautious candidate is:

$$\text{sgn}(d \tau / d \lambda) = \text{sgn}(L^\mu \nabla_\mu \omega)$$

This says that the local orientation of time-like ordering is tied to how a light-like or radiation-flow direction samples the phase-order gradient. In plain language: time orientation follows the sign of light-flow through the phase structure.

This should be treated as a hypothesis to be derived or rejected, not as a finished law.

4. Standard limit

The strongest version of this target must recover ordinary physics when the RUT bridge decouples. The required correspondence condition is:

$$B(a) \rightarrow 0 \text{ implies } G_{mn} \rightarrow 8 \pi G T_{mn}^m$$

or, in cosmological language:

$$B(a) \rightarrow 0 \text{ implies GR / LambdaCDM correspondence limit}$$

This limit is essential. Without it, RUT would risk contradicting the many places where GR and standard cosmology already work extremely well.

5. Main derivation challenge

The key technical challenge is to derive the mixed term $T_{mn}(\omega L)$ from an action, variational principle, or effective-field scaffold. The term should not remain only symbolic. It needs a mathematical origin, conservation bookkeeping, stability conditions, and observational consequences.

Target action direction:

$$S = \int d^4x \sqrt{-g} \left[M_{Pl}^2 R/2 + L_m + L_\omega + L_L + \xi B(a) L_{mix}(\omega, L^\mu, g_{mn}) \right]$$

A future derivation attempt should vary this action with respect to g_{mn} , ω , and the light-flow/null-structure variables, then check whether the resulting equations preserve conservation, no-ghost stability, and the GR limit.

6. Falsifier and success condition

Falsifier: If no stable, conserved, GR-compatible mixed term can be defined, then the geometry-light-time coupling should remain a metaphor or be removed from the physics-facing core.

Success condition: RUT becomes substantially stronger if the mixed term can be derived in a way that predicts a measurable signature - for example in BAO/ r_d behavior, gravitational-wave H_0 inference, phase-state waveform signatures, or scale-dependent growth/lensing residuals.

7. Suggested next work package

Next derivation attempt: define ω precisely, choose whether L^μ is a null vector, radiation-flow field, or effective current, write the simplest allowed L_{mix} , vary the action, and test whether the result reduces continuously to GR when $B(a) \rightarrow 0$.

Suggested filename

RUT_Derivation_Target_Note_Geometry_Light_Time_Coupling_v1.pdf